Kevin K. Yang

Protein engineering · Deep generative models · Biological sequence modeling

Microsoft, One Memorial Drive, Cambridge, MA, 02140

kvkyang@gmail.com | www.yangkky.github.io | Google Scholar | @KevinKaichuang

Summary

Kevin K. Yang's research focuses on developing machine learning methods to design and understand proteins. Specifically, his work aims to engineer novel and complex functions by modeling proteins as dynamic, context-mediated biomolecules. He is currently a principal researcher in the BioML group at Microsoft Research. Kevin earned his PhD in chemical engineering from the California Institute of Technology, and a BS in Chemical and Biomolecular Engineering from The Ohio State University.

Education

California Institute of Technology *Pasadena, CA*

PhD Chemical Engineering *July 2014 - Dec 2018*

- Advisor: Dr. Frances Arnold
- Thesis: Probabilistic Protein Engineering

The Ohio State University *Columbus, OH*

BS Chemical and Biomolecular Engineering *September 2007 - June 2011*

Professional Experience

2024-present	**Principal Researcher**, Microsoft Research
2020-2024	**Senior Researcher**, Microsoft Research
2019-2020	**Machine Learning Scientist**, Generate Biomedicines
2018	**Intern**, Ambry Genetics
2015-2017	**Graduate Teaching Assistant**, California Institute of Technology
2011-2014	**Teacher**, Animo Inglewood Charter High School (AICHS)

Publications

Selected Publications

FLIP2: Expanding Protein Fitness Landscape Benchmarks for Real-World Machine Learning Applications. Kieran Didi, Sarah Alamdari, Alex X Lu, Bruce Wittmann, Kadina E Johnston, Ava P Amini, Ali Madani, Maya Czeneszew, Christian Dallago, **Kevin K Yang**. *International Conference on Machine Learning* 2026. 10.64898/2026.02.23.707496.

The Dayhoff Atlas: scaling sequence diversity for improved protein generation. **Kevin K Yang**, Sarah Alamdari, Alex J Lee, Kaeli Kaymak-Loveless, Samir Char, Garyk Brixi, Carles Domingo-Enrich, Chentong Wang, Suyue Lyu, Nicolo Fusi, Neil Tenenholtz, Ava P Amini. Preprint, 2025. 10.1101/2025.07.21.665991.

Computational scoring and experimental evaluation of enzymes generated by neural networks. Sean R Johnson, Xiaozhi Fu, Sandra Viknander, Clara Goldin, Sarah Monaco, Aleksej Zelezniak, **Kevin K Yang**. *Nature Biotechnology*, 2024. 10.1038/s41587-024-02214-2.

Protein structure generation via folding diffusion. Kevin E Wu, **Kevin K Yang**, Rianne van den Berg, Sarah Alamdari, James Y Zou, Alex X Lu, Ava P Amini. *Nature Communications*, 2024. 10.1038/s41467-024-45051-2.

Machine learning-guided channelrhodopsin engineering enables minimally-invasive optogenetics. Claire N Bedbrook, **Kevin K Yang**, J Elliott Robinson, Elisha D Mackey, Viviana Gradinaru, Frances H Arnold. *Nature Methods*, 2019. 10.1038/s41592-019-0583-8. [pdf].

Peer-Reviewed Articles

39. Scalable and cost-efficient custom gene library assembly from oligopools. Chase R Freschlin, **Kevin K Yang**, Philip A Romero. Accepted at *Science Advances*, 2026. [Preprint].

38. Scaling Laws and Architectural Frontiers in Metagenomic Foundation Models. Geraldene Munsamy, Gavin Ayres, Jérémie DONA, Carla Greco, Daniel Anderson, Srijani Sridhar, William Chow, Aaron W Kollasch, Robert J. Pecoraro, Tanggis Bohnuud, Keith Kam, Gus Minto-Cowcher, Marcus H Y Leung, Hassan Sirelkhatim, John St. John, Ali Taghibakhshi, Tyler C. Shimko, Jared Wilber, Timur Rvachov, Saee Gopal Paliwal, Eddie Calleja, Noelia Ferruz, **Kevin K Yang**, Francesco Farina, Philipp Lorenz. Accepted at *International Conference on Machine Learning* 2026.

37. FLIP2: Expanding Protein Fitness Landscape Benchmarks for Real-World Machine Learning Applications. Kieran Didi, Sarah Alamdari, Alex X Lu, Bruce Wittmann, Kadina E Johnston, Ava P Amini, Ali Madani, Maya Czeneszew, Christian Dallago, **Kevin K Yang**. Accepted at *International Conference on Machine Learning* 2026. Preprint.

36. Predicting evolutionary rate as a pretraining task improves genome language model representations. Micaela Elisa Consens, **Kevin K Yang**, Jimmy Hall, Ashley Mae Conard, Bo Wang, Lorin Crawford, Alan Moses, Alex X Lu. Accepted at *International Conference on Machine Learning* 2026. Preprint.

35. Sequence Display enables large-scale sequence–activity datasets for rapid protein evolution. Linqi Cheng, Xinzhe Zheng, Shiyu Jason Jiang, Yu Hu, Yijie Liu, Kaiqiang Yang, Jinyan Rui, Haoxue Ding, Mengxi Zhang, Teng Yuan, Qianglan Lu, Haoxin Ye, Chen-Long Li, Yiming Guo, Zuotong Tian, Anna Qin, Boyang Zhou, **Kevin K Yang**, Xiongyi Huang, Han Xiao. *Nature Biotechnology*, 2026. 10.1038/s41587-026-03087-3.

34. Closing the loop: Experimentally validated methods in artificial intelligence–driven protein design. Clayton W Kosonocky, Sarah Alamdari, **Kevin K Yang**, Ava P Amini. *Current Opinion in Structural Biology*, 2026. 10.1016/j.sbi.2026.103272.

33. Trainable subnetworks reveal insights into structure knowledge organization in protein language models. Ria Vinod, Ava P Amini, Lorin Crawford, **Kevin K Yang**. *PLOS Computational Biology*, 2026. 10.1371/journal.pcbi.1013925.

32. Deep learning guided design of protease substrates. Carmen Martin-Alonso, Sarah Alamdari, Tahoura S Samad, **Kevin K Yang**, Sangeeta N Bhatia, Ava P Amini. *Nature Communications*, 2026. 10.1038/s41467-025-67226-1.

31. Democratizing protein language model training, sharing and collaboration. Jin Su, Zhikai Li, Chenchen Han, Yuyang Zhou, Yan He, Junjie Shan, Xibin Zhou, Xing Chang, Shiyu Jiang, Dacheng Ma, OPMC, Martin Steinegger, Sergey Ovchinnikov, Fajie Yuan. *Nature Biotechnology, 2026*. 10.1038/s41587-025-02859-7.

30. Recent advances in engineering microbial lipases for industrial applications. Geng Wang, Asma Abdella, Mohamadali Fakhari, Jie Dong, **Kevin K Yang**, Shang-Tian Yang. *Biotechnology Advances*, 2025. 10.1016/j.biotechadv.2025.108658. [pdf].

29. Annotating the microbial dark matter with HiFi-NN. Gavin Ayres, Geraldene Munsamy, Michael Heinzinger, Noelia Ferruz, **Kevin K Yang**, Bastiaan Bergman, Philipp Lorenz. *iScience*, 2025. 10.1016/j.isci.2025.112480.

28. Tokenized and continuous embedding compressions of protein sequence and structure. Amy X Lu, Wilson Yan, **Kevin K Yang**, Vladimir Gligorijevic, Kyunghyun Cho, Pieter Abbeel, Richard Bonneau, Nathan C Frey. *Patterns*, 2025. 10.1016/j.patter.2025.101289.

27. ProtNote: a multimodal method for protein–function annotation. Samir Char, Nathaniel Corley, Sarah Alamdari, **Kevin K Yang**, Ava P Amini. *Bioinformatics*, 2025. 10.1093/bioinformatics/btaf170.

26. Toward deep learning sequence–structure co-generation for protein design. Chentong Wang, Sarah Alamdari, Carles Domingo-Enrich, Ava P Amini, **Kevin K Yang**. *Current Opinion in Structural Biology*, 2025. 10.1016/j.sbi.2025.103018. [pdf].

25. Benchmarking uncertainty quantification for protein engineering. Kevin P Greenman, Ava P Amini, **Kevin K Yang**. *PLOS Computational Biology*, 2025. 10.1371/journal.pcbi.1012639.

24. Computational scoring and experimental evaluation of enzymes generated by neural networks. Sean R Johnson, Xiaozhi Fu, Sandra Viknander, Clara Goldin, Sarah Monaco, Aleksej Zelezniak, **Kevin K Yang**. *Nature Biotechnology*, 2024. 10.1038/s41587-024-02214-2.

23. Convolutions are competitive with transformers for protein sequence pretraining. **Kevin K Yang**, Nicolo Fusi, Alex X Lu. *Cell Systems*, 2024. 10.1016/j.cels.2024.01.008.

22. Feature reuse and scaling: Understanding transfer learning with protein language models. Francesca-Zhoufan Li, Ava P Amini, Yisong Yue, **Kevin K Yang**, Alex X Lu. *International Conference on Machine Learning* 2024. [PMLR].

21. Protein structure generation via folding diffusion. Kevin E Wu, **Kevin K Yang**, Rianne van den Berg, Sarah Alamdari, James Y Zou, Alex X Lu, Ava P Amini. *Nature Communications*, 2024. 10.1038/s41467-024-45051-2.

20. Machine learning for protein engineering. Kadina E Johnston, Clara Fannjiang, Bruce J Wittmann, Brian L Hie, **Kevin K Yang**, Zachary Wu. *Machine Learning in Molecular Sciences*, 2023. 10.1007/978-3-031-37196-7_9. [pdf].

19. Deep self-supervised learning for biosynthetic gene cluster detection and product classification. Carolina Rios-Martinez, Nicholas Bhattacharya, Ava P Amini, Lorin Crawford, **Kevin K Yang**. PLOS computational biology, 2023. 10.1371/journal.pcbi.1011162.

18. Masked inverse folding with sequence transfer for protein representation learning. **Kevin K Yang**, Niccolò Zanichelli, Hugh Yeh. *Protein Engineering, Design and Selection*, 2022. 10.1093/protein/gzad015.

17. Exploring evolution-based &-free protein language models as protein function predictors. Mingyang Hu, Fajie Yuan, **Kevin K. Yang**, Fusong Ju, Jin Su, Hui Wang, Fei Yang, Qiuyang Ding. *NeurIPs* 2022. [pdf].

16. Randomized gates eliminate bias in sort-seq assays. Brian . Trippe, Buwei Huang, Erika A. DeBenedictis, Brian Coventry, Nicholas Bhattacharya, **Kevin K Yang**, David Baker, Lorin Crawford. *Protein Science*, 2022. 10.1002/pro.4401.

15. Evolutionary velocity with protein language models. Brian L Hie, **Kevin K Yang**, Peter S Kim. *Cell Systems*, 2022. 10.1016/j.cels.2022.01.003.

14. Machine learning modeling of family wide enzyme-substrate specificity screens. Samuel Goldman, Ria Das, **Kevin K Yang**, Connor W Coley. PLoS computational biology, 2022. 10.1371/journal.pcbi.1009853.

13. A topological data analytic approach for discovering biophysical signatures in protein dynamics. Wai Shing Tang, Gabriel Monteiro da Silva, Henry Kirveslahti, Erin Skeens, Bibo Feng, Timothy Sudijono, **Kevin K Yang**, Sayan Mukherjee, Brenda Rubenstein, Lorin Crawford. *PLoS computational biology*, 2022. 10.1371/journal.pcbi.1010045.

12. Adaptive machine learning for protein engineering. Brian L Hie and **Kevin K Yang**. *Current Opinion in Structural Biology*, 2022. 10.1016/j.sbi.2021.11.002.

11. FLIP: Benchmark tasks in fitness landscape inference for proteins. Christian Dallago, Jody Mou, Kadina E Johnston, Bruce J Wittmann, Nicholas Bhattacharya, Samuel Goldman, Ali Madani, **Kevin K Yang**. *NeurIPS 2021 Datasets and Benchmarks Track*. 10.1101/2021.11.09.467890.

10. Protein sequence design with deep generative models. Zachary Wu, Kadina E Johnston, Frances H Arnold, **Kevin K Yang**. *Current Opinion in Chemical Biology*, 2021. 10.1016/j.cbpa.2021.04.004.

9. Learned embeddings from deep learning to visualize and predict protein sets. Christian Dallago, Konstantin Schütze, Michael Heinzinger, Tobias Olenyi, Maria Littmann, Amy X Lu, **Kevin K Yang**, Seonwoo Min, Sungroh Yoon, James T Morton, Burkhard Rost. *Current Protocols*, 2021. 10.1002/cpz1.113.

8. Signal Peptides Generated by Attention-Based Neural Networks. Zachary Wu, **Kevin K Yang**, Michael J Liszka, Alycia Lee, Alina Batzilla, David Wernick, David P. Weiner, Frances H Arnold. *ACS Synthetic Biology*, 2020. 10.1021/acssynbio.0c00219.

7. Machine learning-guided channelrhodopsin engineering enables minimally-invasive optogenetics. Claire N Bedbrook, Kevin K Yang, J Elliott Robinson, Elisha D Mackey, Viviana Gradinaru, Frances H Arnold. *Nature Methods*, 2019. 10.1038/s41592-019-0583-8.

6. Machine-learning-guided directed evolution for protein engineering. Kevin K Yang, Zachary Wu, Frances H Arnold. *Nature Methods*, 2019. 10.1038/s41592-019-0496-6.

5. Batched stochastic Bayesian optimization via combinatorial constraints design. **Kevin K Yang**, Yuxin Chen, Alycia Lee, Yisong Yue. *AIStats* 2019. [pdf].

4. The Generation of Thermostable Fungal Laccase Chimeras by SCHEMA-RASPP Structure-Guided Recombination in Vivo. Ivan Mateljak, Austin Rice, **Kevin K Yang**, Thierry Tron, Miguel Alcalde. *ACS Synthetic Biology*, 2019. 10.1021/acssynbio.8b00509.

3. Learned protein embeddings for machine learning. **Kevin K Yang**, Zachary Wu, Claire N Bedbrook, France H Arnold. *Bioinformatics*, 2018. 10.1093/bioinformatics/bty178.

2. Machine learning to predict eukaryotic expression and plasma membrane localization of engineered integral membrane proteins. Claire N Bedbrook*, **Kevin K Yang***, Austin J Rice, Viviana Gradinaru, Frances H Arnold. *PLOS Computational Biology*, 2017. 10.1371/journal.pcbi.1005786.

1. Structure-Guided SCHEMA Recombination Generates Diverse Chimeric Channelrhodopsins. Claire N Bedbrook, Austin J Rice, **Kevin K Yan**g, Xiaozhe Ding, Siyuan Chen, Emily M LeProust, Viviana Gradinaru, Frances H Arnold. *Proceedings of the National Academy of Sciences*, 2017. 10.1073/pnas.1700269114.

Preprints

10. Evolutionary conditioning enables guided generation of functionally diverse enhancers. Andrew G Duncan, Micaela E Consens, Lorin Crawford, Jennifer A Mitchell, Alan M Moses, **Kevin K Yang**, Alex X Lu. Preprint, 2026. 10.64898/2026.04.13.718170.

9. Designing AI-programmable therapeutics with the EDEN family of foundation models. Geraldene Munsamy, Gavin Ayres, Carla Greco, Keith Kam, Gus Minto-Cowcher, John John St, Tanggis Bohnuud, Matthew Bakalar, William Chow, Robert Pecoraro, Marcelo DT Torres, Aaron Kollasch, Marcus Leung, Hassan Sirelkhatim, Francesco Farina, Connor McGinnis, Srijani Sridhar, Daniel Anderson, Francesco Oteri, Ali Taghibakhshi, Jeremie Dona, Tyler Shimko, Cedric Steenbeke, Alexandros Papadopoulos, Malcolm Krolick, Fabian Spoendlin, Purba Gupta, Sandeep Kumar, Anne Bara, Jared Wilbur, Noelia Ferruz, Timur Rvachov, Fangping Wan, Hanqun Cao, Hyun-Su Lee, Japan Mehta, Raphael Chaleil, Valerio Pereno, Sid Potti, Chris Emerson, Roy Tal Dew, **Kevin K Yang**, Eric Nguyen, Neha Tadimeti, Jillian F Banfield, Alicia Frame, Emma Bolton, David Ruau, Rory Kelleher, Anthony Costa, Kimberley Powell, Cesar de la Fuente-Nunez, Glen-Oliver Gowers, Oliver Vince, Jonathan Finn, Philipp Lorenz. Preprint, 2026. 10.64898/2026.01.12.699009.

8. Few-shot Protein Fitness Prediction via In-context Learning and Test-time Training. Felix Teufel, Aaron W Kollasch, Yining Huang, Ole Winther, **Kevin K Yang**, Pascal Notin, Debora S Marks. Preprint, 2025. 2512.02315.

7. The Dayhoff Atlas: scaling sequence diversity for improved protein generation. **Kevin K Yang**, Sarah Alamdari, Alex J Lee, Kaeli Kaymak-Loveless, Samir Char, Garyk Brixi, Carles Domingo-Enrich, Chentong Wang, Suyue Lyu, Nicolo Fusi, Neil Tenenholtz, Ava P Amini. Preprint, 2025. 10.1101/2025.07.21.665991.

6. MotifBench: A standardized protein design benchmark for motif-scaffolding problems. Zhuoqi Zheng, Bo Zhang, Kieran Didi, **Kevin K Yang**, Jason Yim, Joseph L Watson, Hai-Feng Chen, Brian L Trippe. Preprint, 2025. 2502.12479.

5. Enzymeflow: Generating reaction-specific enzyme catalytic pockets through flow matching and co-evolutionary dynamics. Chenqing Hua, Yong Liu, Dinghuai Zhang, Odin Zhang, Sitao Luan, Kevin K Yang, Guy Wolf, Doina Precup, Shuangjia Zheng. Preprint, 2024. 2410.00327.

4. Generating all-atom protein structure from sequence-only training data. Amy X Lu, Wilson Yan, Sarah A Robinson, **Kevin K Yang**, Vladimir Gligorijevic, Kyunghyun Cho, Richard Bonneau, Pieter Abbeel, Nathan Frey. Preprint, 2024. 10.1101/2024.12.02.626353.

3. Mapping the combinatorial coding between olfactory receptors and perception with deep learning. Seyone Chithrananda, Judith Amores, **Kevin K Yang**. Preprint, 2024. 10.1101/2024.09.16.613334.

2. Sequence vs. Structure: delving deep into data-driven protein function prediction. Xiaochen Tian, Ziyin Wang, **Kevin K Yang**, Jin Su, Hanwen Du, Qiuguo Zheng, Guibing Guo, Min Yang, Fei Yang, Fajie Yuan. Preprint, 2023. 10.1101/2023.04.02.534383.

1. Protein generation with evolutionary diffusion: sequence is all you need. Sarah Alamdari, Nitya Thakkar, Rianne van den Berg, Alex Xijie Lu, Nicolo Fusi, Ava Pardis Amini, **Kevin K Yang**. Preprint, 2023. 10.1101/2023.09.11.556673.

Patents

4. Joint prediction of odorant-olfactory receptor binding and odorant percepts. Judith Amores, Seyone Chithrananda, **Kevin K Yang**. Patent pending. US Patent App. 18/375,219. Filed 2025.

3. Unbiased sorting and sequencing of objects via randomized gating schemes. Brian L Trippe, Lorin Crawford, **Kevin K Yang**, Nicholas Bhattacharya. US Patent no. 12571733. Issued 2026.

2. Optimizing proteins using model based optimizations. **Kevin K Yang**, Jacob D Feala, Maxim Baranov, Brinda Monian. US Patent no. 12020776. Issued 2024.

1. Engineered light-sensitive proteins. Viviana Gradinaru, Claire N Bedbrook, Frances H Arnold, **Kevin K Yang**. US Patent no. 18517703. Issued 2024.

Selected Software & Open Source

4. FLIP/FLIP2: Community benchmark for protein fitness landscape prediction. 2021/2026. [Website].

3. Dayhoff Atlas: Models and datasets for generating protein sequences. 2025. [Code (Github)] [Models and datasets (HuggingFace)].

2. EvoDiff: Discrete diffusion framework for unconditional and conditional protein sequence generation. 2023. [Code (Github)] [Model weights (Zenodo)].

1. Protein Sequence Models: protein sequence modeling including I/O tools and pretrained convolutional models. 2022. [Code (Github)] [CARP model weights (Zenodo)].

Awards & Honors

2017 **Teaching Assistantship Award**, Caltech Chemistry and Chemical Engineering
2015-2018 **Biotechnology Leadership Program**, Caltech Rosen Center
2011 **Graduate Research Fellowship**, National Science Foundation

Teaching Experience

2018	**Introduction to biomolecular engineering**, Teaching Assistant	*Caltech*
2016	**Introduction to biomolecular engineering**, Teaching Assistant	*Caltech*
2015	**Chemical reaction engineering**, Teaching Assistant	*Caltech*
2011-2014	**Algebra 1**, Teacher	*AICHS*
2011-2014	**Physics**, Teacher	*AICHS*

Mentoring

Thesis Committees

2023-2026	**Ria Vinod**, Center for Computational and Molecular Biology, Advisor: Lorin Crawford.	*Brown University*
2023-2025	**Chase Freschlin**, Biochemistry, Advisor: Philip Romero	*Wisconsin University*
2023-2025	**Francesca Zhoufan-Li**, Bioengineering, Advisors: Frances Arnold and Yisong Yue.	*Caltech*

Interns

2026 Anna Sappington, Jimena Olivares Rivera, Khue Tran
2025 Clayton Kosonocky
2024 Sidhanth Holalkere, Suyue Lyu, Chentong Wang
2023 Seyone Chithrananda, Michael Wornow, Ekene Ezeunala, Divya Nori, Cydney Martell
2022 Francesca Zhoufan-Li, Kevin Wu, Amy Wang, Nitya Thakkar
2021 Nicholas Bhattacharya, Carolina Rios-Martinez, Kevin P. Greenman
2020 Soumya Ram

Research

2026 Benjamin Perry, Kasie Leung
2024 Da (Darren) Xu, Kaeli Kaymak-Loveless, Alex J. Lee, Garyk Brixi, Amelia Zug, Nathaniel Corley, Samir Char
2023-2026 Zhiqing Xu, Rana Barghout, Sevahn Vorperian
2021-2025 Amy X. Lu
2021 Peishan Huang, Jody Mou
2020 Hugh Yeh
2016-2019 Alycia Lee

Professional Activities

2026	**Organizing Committee**, New England Computational Biology Day
2024-2026	**Area Chair**, ICML, NeurIPS, ICLR
2021-2022	**Founder and Organizer**, ML for Protein Engineering Seminar Series
2019-2026	**Ad hoc reviewer**, ICML, ICLR, NeurIPS, *Cell*, *Science*, *Bioinformatics*, *Nature*, *Genome Biology*, *ACS Catalysis*, *Nature Methods*, *ACS Synthetic Biology*, *Nature Communications*, *Nature Chemical Biology*, *PLOS Computational Biology*, *PNAS*, *JACS*, *iScience*

Grants & Funding

Under consideration

EvoPlast: Sequence-display guided machine learning for the engineering of symbiotic plastics. DOE Genesis Mission. Unfunded co-PI. 2026.

AI-coupled continuous evolution for scalable protein biodesign. DOE Genesis Mission. Unfunded co-PI. 2026.

Presentations

Invited Talks

35. Nvidia. Virtual talk. May 2026.

34. Session on *Advances in Experimental Methods for Protein Engineering*. American Institute of Chemical Engineers Annual Meeting. Boston, MA. November 2025.

33. Cell Circuits and Epigenomics Seminar Series. Broad Institute. Cambridge, MA. June 2025.

32. Arena Bio. Cambridge, MA. October 2024.

31. Quantitative Biodesign Symposium. Duke University. Durham, NC. October 2024.

30. MIT Bioinformatics Seminar. Cambridge, MA. October 2024.

29. ICML AI4Science Workshop. Vienna, Austria. July 2024.

28. Wenner-Gren Symposium: *Embracing the Alphafold Revolution: Unveiling the Future of Structutal Biology*. Stockholm, Sweden. May 2024.

27. EMBL-EBI Industry Programme workshop on *Advances in machine learning for protein design*. Wellcome Research Center. Hinxton, UK. May 2024.

26. 5.64/HST.539 of Frontiers of Interdisciplinary Science in Human Health and Disease. MIT. Cambridge, MA. April 2024.

25. New England Biolabs, Ipswich, MA. April 2024.

24. RosettaCon, Cambridge, MA. March 2024.

23. Google Brain. Cambridge, MA. October 2023.

22. *ML Protein Engineering* Seminar Series. Virtual talk. September 2023.

21. International Conference on Biological Physics. Seoul, Korea. August 2023.

20. Gordon Conference on Computer-Assisted Drug Discovery. Mt. Snow, Vermont. July 2023.

19. M2D2, Valence Labs. Virtual talk. June 2023.

18. BE/BI/205: Deep Learning for Biological Data. Caltech. Pasadena, CA. May 2023.

17. Conference on *Interplay between AI and mathematical modelling in the post-structural genomics era*. CIRM, Marseilles, France. March 2023.

16. Nvidia GTC. Virtual talk. March 2023.

15. Cell & Systems Biology Departmental Seminar. University of Toronto. Toronto, Canada. February 2023.

14. Biomedical Informatics Seminar, Columbia University. New York, NY. October 2022.

13. Bioinforange Conference. October 2022.

12. Merck. Virtual talk. August 2022.

11. Journal Club, OpenBioML. Virtual talk. May 2022.

10. Genomics Seminar, UC Riverside. Virtual talk. May 2022.

9. AI for Drug Discovery Seminar, University of Washington. Virtual talk. May 2022.

8. Bioinformatics Student Symposium, Boston University. Boston, MA. May 2022.

7. Molecular Maker Lab Institute Conference, UIUC. Urbana-Champaign, IL. March 2022.

6. AI4Science Seminar, Chalmers University. Virtual talk. March 2022.

5. Machine learning seminar, IBM. Virtual talk. December 2021.

4. Protein Engineering Congress Global. Virtual talk. October 2021.

3. GSK Data Forum. February 2021.

2. Janelia Research Campus. Ashburn, VA. May 2019.

1. Gray-Hill Seminar Series, Occidental College. Los Angeles, CA. June 2018.